Merchant Silicon in Carrier Networks

November 1, 2022

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Agenda

- Introduction
- DNX Portfolio and Cadence
- Partners & Ecosystem
- DNX in Access Edge and Core
- DNX Architecture Efficiency
- DNX DDC for AI / ML
- Summary

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Broadcom Merchant Silicon Switch Portfolio

Enterprise



Trident

Hyperscale Data Center



Tomahawk

Service Provider



Jericho / Qumran
Ramon

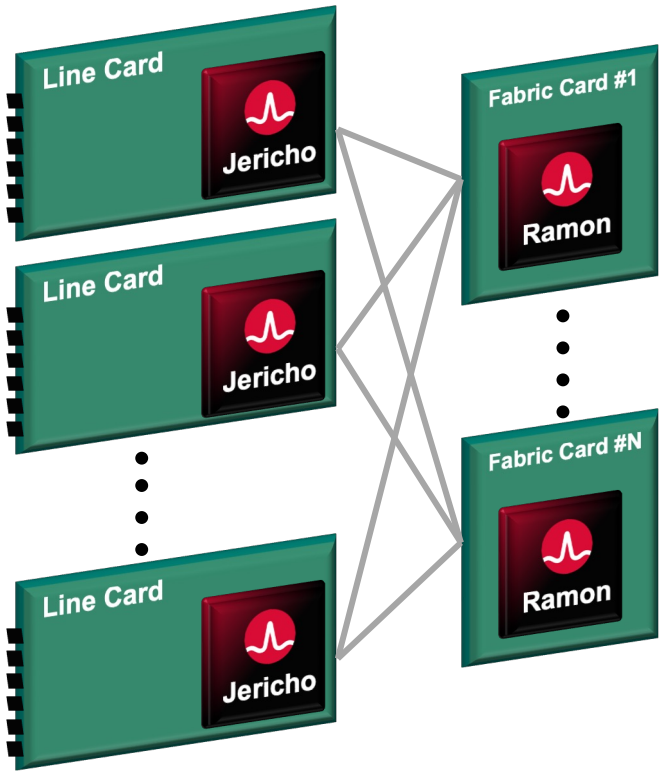
Domain-Specific Switching Silicon

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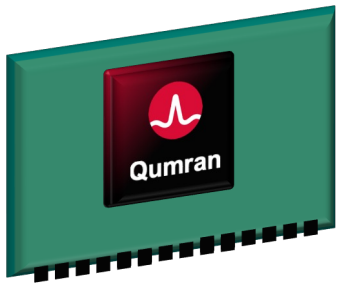
DNX – Deployed In All Networks

Jericho & Ramon
Distributed / Chassis



- Non-blocking switching
- Controlled latency
- Deep packet buffering (lossless)
- Large-scale Traffic Management
- Programmable PP
- Millions of IP routes
- Redundancy and protection
- Unified silicon architecture & SDK

Qumran
Centralized / Fixed

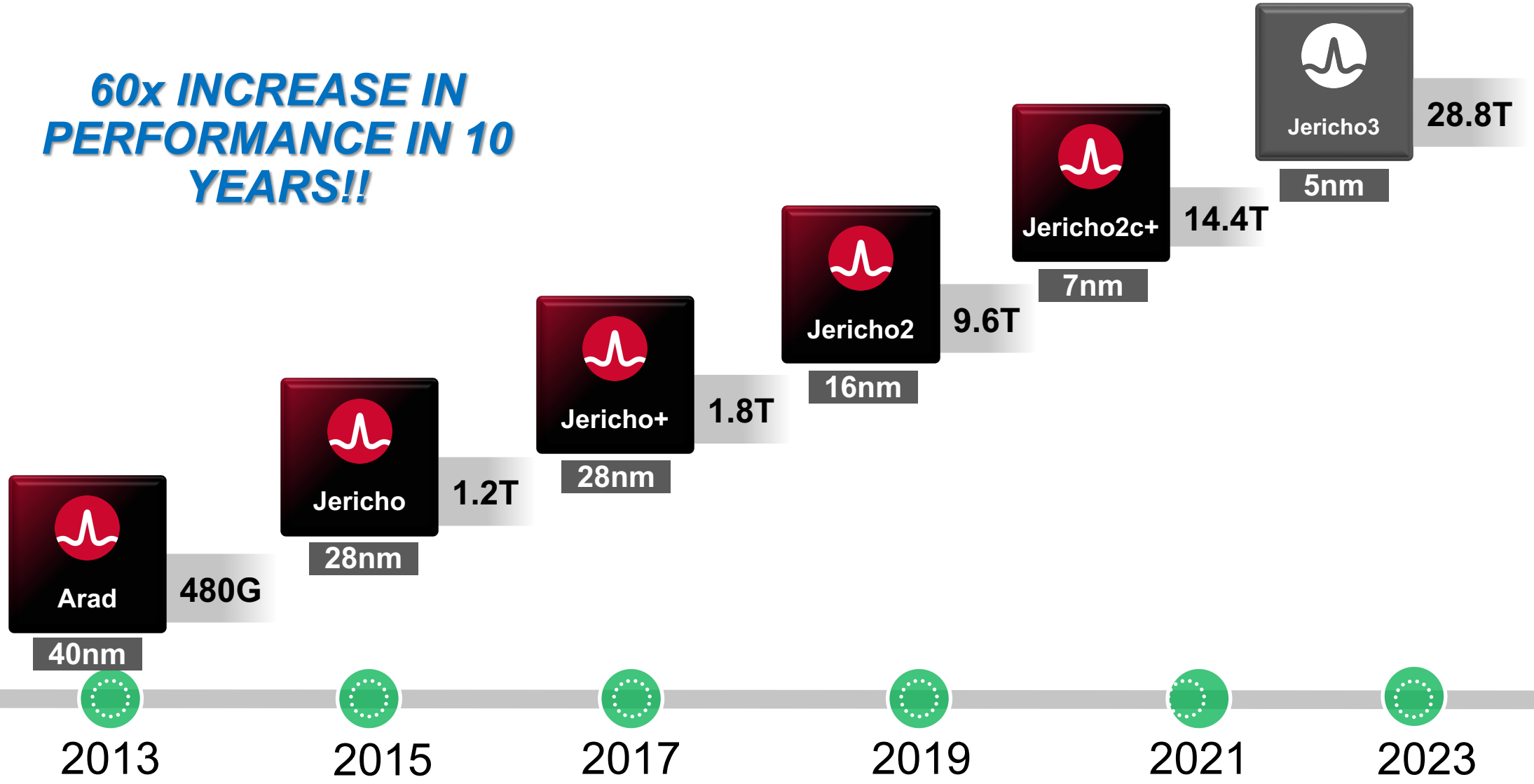


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Merchant Silicon Jericho Innovation Cadence

60x INCREASE IN PERFORMANCE IN 10 YEARS!!

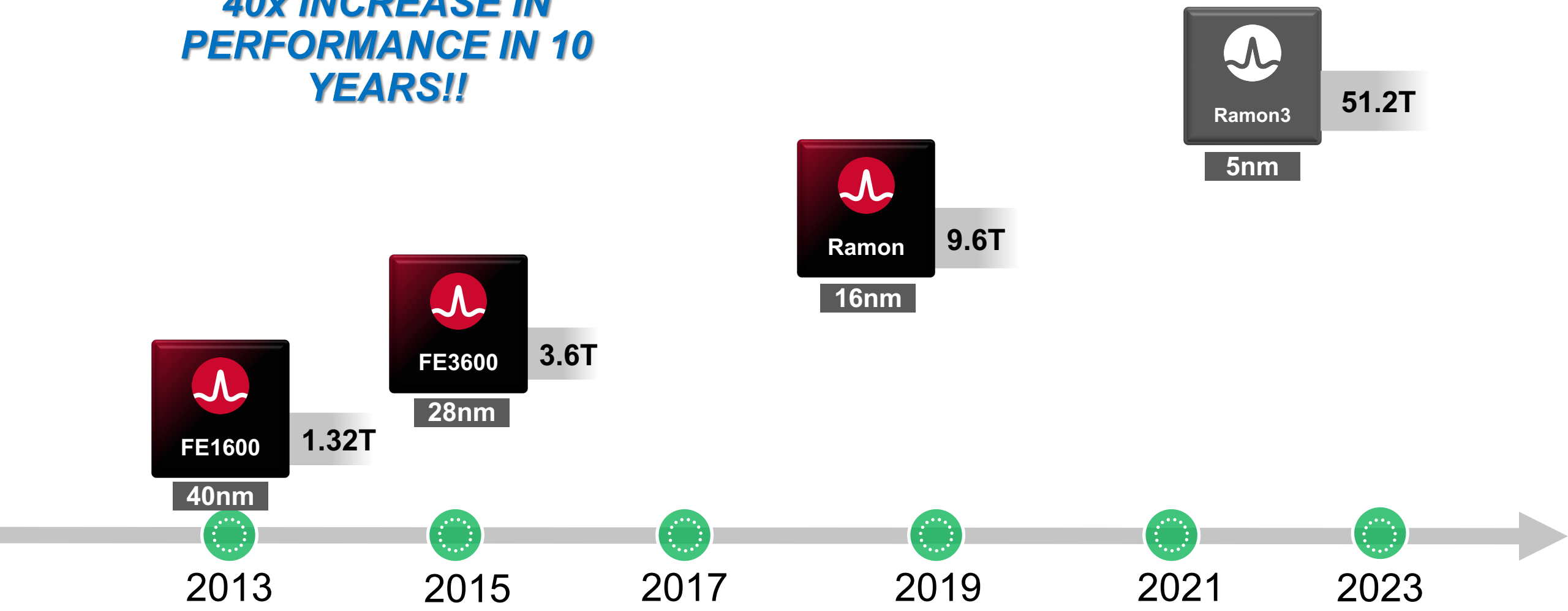


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Merchant Silicon Ramon Innovation Cadence

40x INCREASE IN PERFORMANCE IN 10 YEARS!!



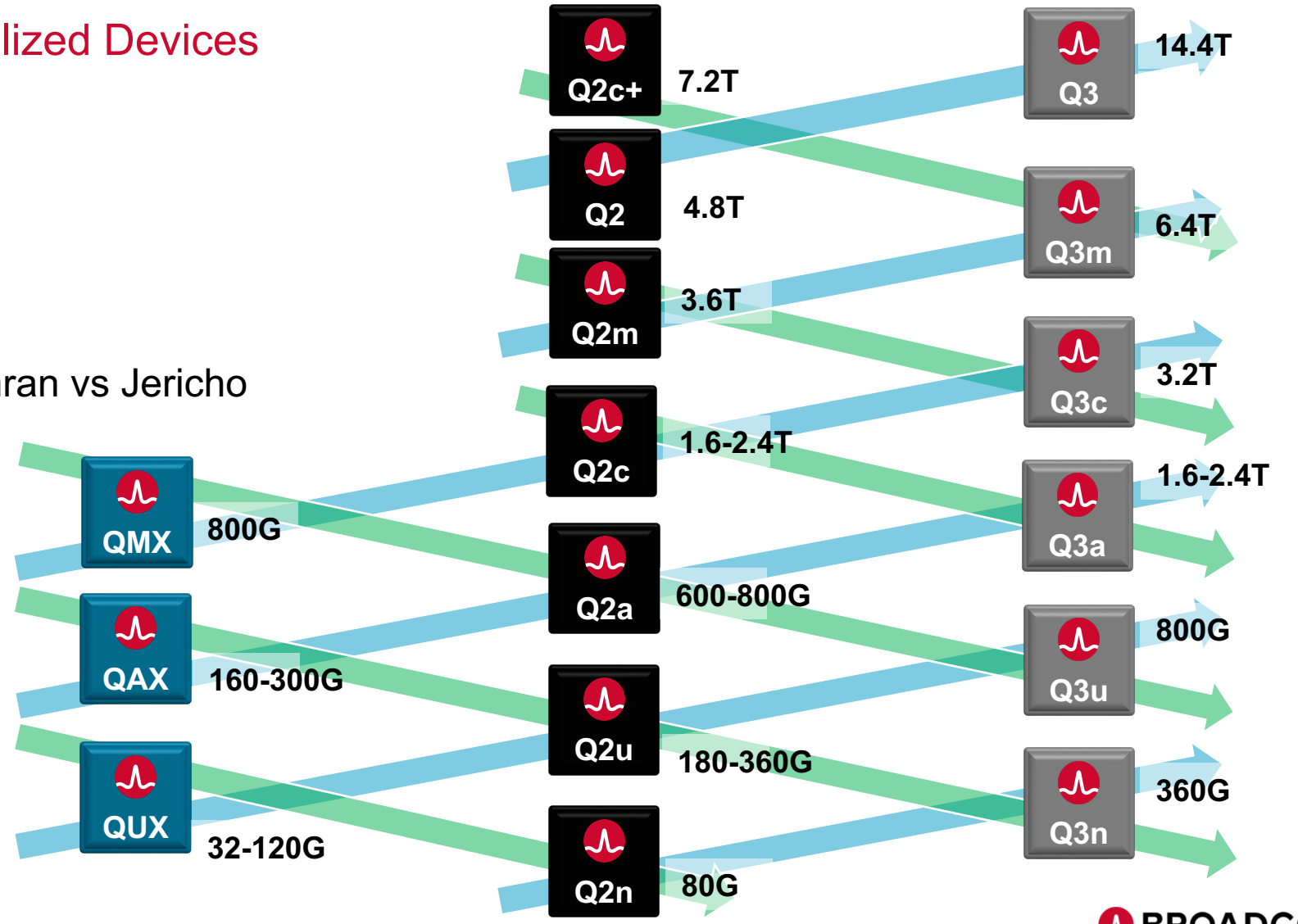
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3 Generations of Qumran in Access, Metro and Edge

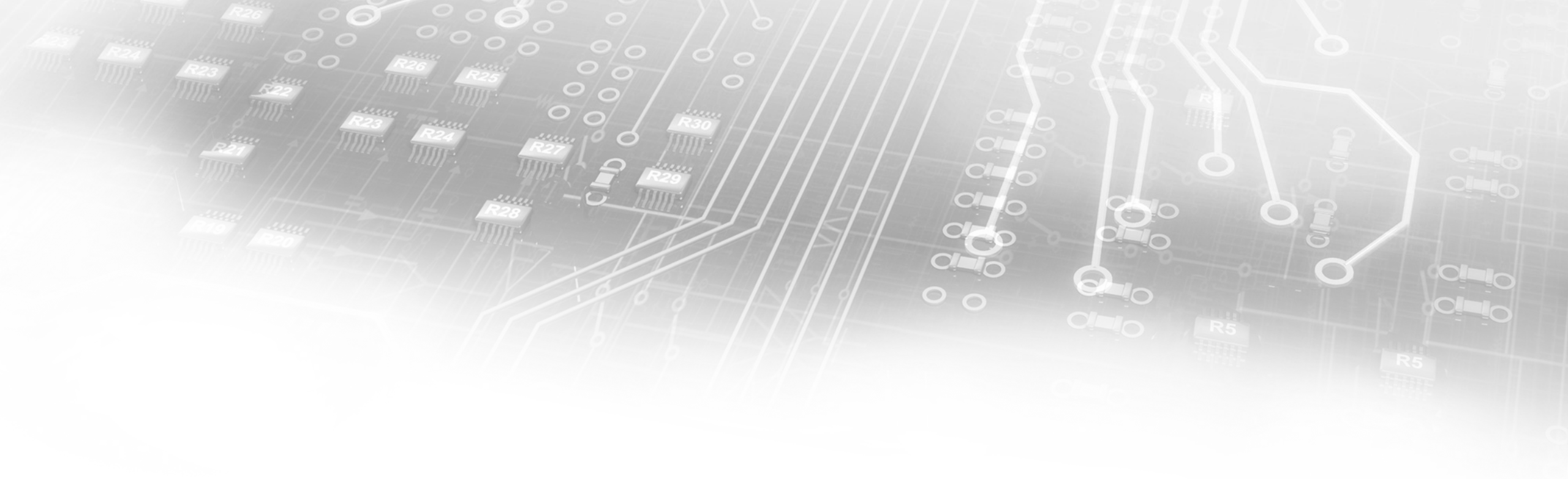
5nm Lineup refresh of all Centralized Devices

- 100G, 50G and 25G lanes
- Line rate MACsec on all ports
- Low power 4W-5W per 100G
- JR3-software compatible
- No feature compromise in Qumran vs Jericho



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Partners & Ecosystem

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Merchant Silicon Enabled Ecosystem

Network Controller

OPEN DAYLIGHT
lagopus
NEC
JUNIPER NETWORKS OPENCONTRAIL

Orchestration & Automation

ca technologies A Broadcom Company
openstack
CHEF
apstra
puppet labs

NV Controller

vmware
midokura
NSX

Disaggregated NOS Options

SONiC
ipinfusion
ARRCUS NETWORK DIFFERENT
infinera
ADVA
exaware
VOLTA Networks
DRIVENETS
rtbrick Network Freedom

ODM HW Partners

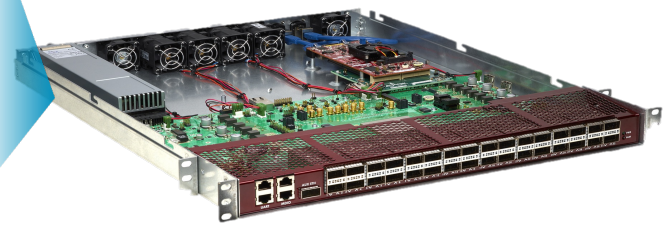
廣達電腦 Quanta Computer
英業達 Inventec
DELTA
ufiSpace
Accton Making Partnership Work
ALPHA Networks
Celestica

Management

NOS

SDK

Tomahawk | Trident | Jericho



OEM SW Partners

ARISTA
JUNIPER NETWORKS
H3C
ciena
Ruijie
FiberHome
HUAWEI
NOKIA
ZTE中兴
DELL
cisco

OEM HW Partners

ARISTA
JUNIPER NETWORKS
H3C
ciena
Ruijie
FiberHome
HUAWEI
NOKIA
ZTE中兴
DELL
cisco

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Merchant Silicon-Based OEM Platforms for Every Network Node



Vendor	Access	Aggregation	Service Edge	Core	DCI	DC Leaf/Spine	DC Top of Rack
ARISTA	7020R	7280R 7500R	7280R 7500R	7800R3 7500R	7800R3 7500R / R7280R	7300X 7500R/E	7260X 7280R/E
ciena	516X	517x			8180		
CISCO	NCS540	ASR900/907 NCS5500/560	NCS5500/NCS5700	NCS5500/NCS5700 NCS6000	NCS5500/NCS5700 NCS6000		
ERICSSON	R6672/R6X71/ R6675	R6675/R6672/ R6273/R6274	SSR8000			HDS8000	HDS8000
HUAWEI	CE6850/55 CE5800	CE8800	CE12800	CE12800	NE9000 NE5000E/40E	CE8800 CE12800	CE6850/55
H3C	S6300/RA5300	S12500R	S12500R	S12500R	S12500	S9800	S6800
Infinera	DRX-30	DRX-90	DRX-90				
JUNIPER NETWORKS	ACX7024 ACX710	ACX5448 ACX7100 ACX7509				QFX5xxx	QFX5xxx
NOKIA	7210 7250-IXR-e	7250-IXR	7250-IXR-R4			7250-IXR-s 7220-IXR 7250 IXR-6e 7250 IXR-10e	7220-IXR Nuage 7850

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OEM NOS + White Box Ecosystem

ciena

Infinera

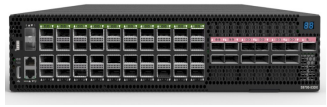
Cisco IOS XR7

JUNOS

ARISTA EOS Extensible Operating System



Metro Aggregation



Metro Edge



Metro Core



DC Spine



DC Leaf

Accton Making Partnership Work

Celestica

DELTA

廣達電腦 Quanta Computer

ufiSpace

QumranMX | Jericho2 | Ramon | Tomahawk | Tomahawk2 | Trident3 | Trident4

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Disaggregated Solution for Every SP Network Node



OLT

QumranAX

REPLY



vBNG

Jericho2c

rtbrick Network Freedom



Metro Aggregation

Jericho2c

ARRCUS NETWORK DIFFERENT



Metro Core and Edge

Jericho2

DRIVENETS™



Cell Site Router

QumranAX

ipinfusion™



Fronthaul Gateway

Monterey

Coming soon

Accton Making Partnership Work

Celestica

DELTA

廣達電腦 Quanta Computer

ufiSpace

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Merchant Silicon-Based ODM Platforms for Every Network Node





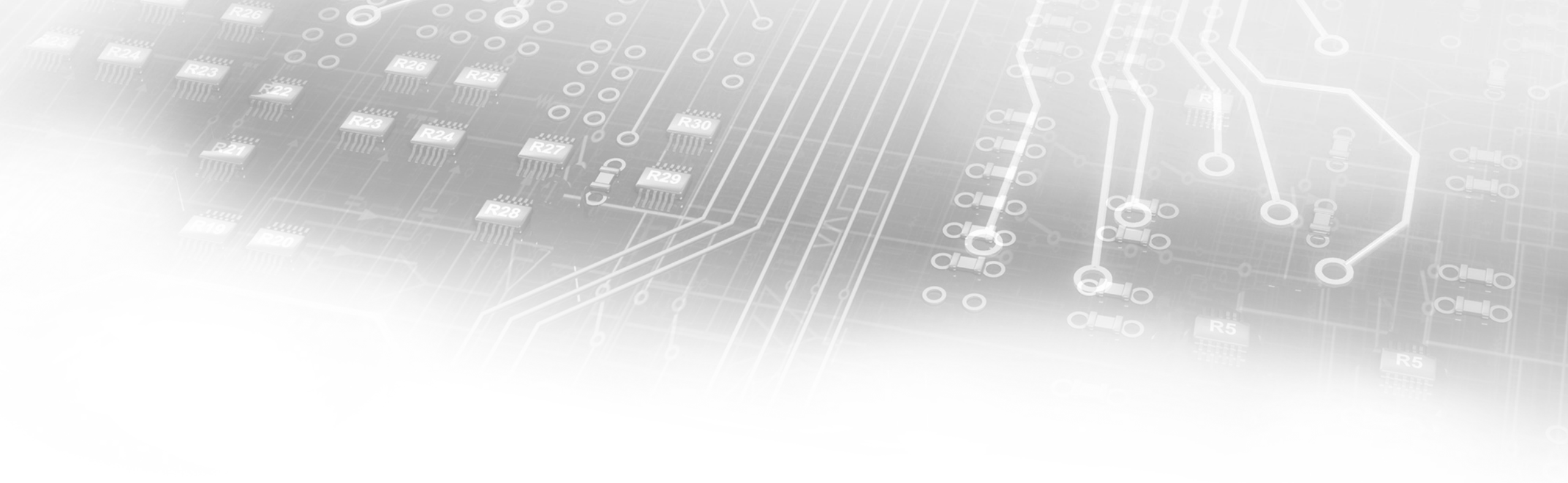




Vendor	Access	Aggregation	Service Edge	Core	DCI	DC Leaf/Spine	DC Top of Rack
Accton Edge-corE NETWORKS	AS5915 AS7315 AS7316	AS7535 AS7525 AS7515	AS5926 AS5916 AS7946	AS7946 AS7936 AS7926	AS7926 AS7826 AS5835	AS973X AS7812 AS972X AS7712 AS971X AS7710	AS7826 AS583X AS7726 AS5812 AS7716 AS4630
ALPHA	STX-60D0	STX-60D0	STX-60x0	STC-60x0	STC-60x0	SNJ-60x0 SNH-60x0	SNJ-61x0 SNH-61x2
Celestica	S3020 S3010	S3020	S3020	D8060 DX510 DX510 DX500	D8060 DX510 DX510 DX500	DX430 DX410 DX400 D7020 D8030 DX010	DX040 DX032 D3040 D3030 DX030 E1062
DELTA	AGC7008S DT-G26X2SL	DT-G26X2SL AGCV208S AGC7648	DT-G32X2SL AGCX424S AGC5648S	DT-V36C4S AGC023S AGCC048	AGCXD40S AGCVA48S AGCC048	DD-S32C8 AG9064 DD-S64C4S AGC032 DD-S32C4S AG5648	DD-T32C2S AGV848 AG6248C AG9032 AG7648 AGV424
Inventec				D8216 D8212	D8216 D8212	D7332 ASL74xx D6432 / D6456	NE1054 D6356 D6332
QCT Quanta CLOUD TECHNOLOGY	T4024	T4032	T4048	T9036 T7080 T9018 T7040 T9010	T9036 T7080 T9018 T7040 T9010	T8128 T7128 T9064 T7064 T9032 T7032	T4048 IX7 T8024 IX8
ufiSpace	G3100 S9501	S9600 S9500 S9510	S9600 S9510	S9700 S9710 S9701	S9700 S9710 S9705	S9300 S9100 S8900	S9310 S8810 S7810

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DNX in Access Edge and Core

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Merchant Silicon In The Service Provider Space

4G/5G Cell-Site Routers (AT&T SIAD, TIP DCSG)

TELECOM INFRA PROJECT



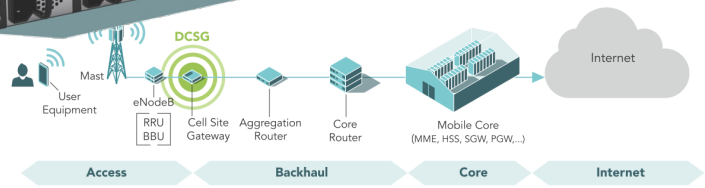
Qumran-UX

n-AX

sdx central

SD-WAN Security 5G AI Edge Data Center IoT SDN Storage NFV Containers APM/NPM Cloud

AT&T's White Box Cell-Site Routers Go Into Production Carrying Live Traffic



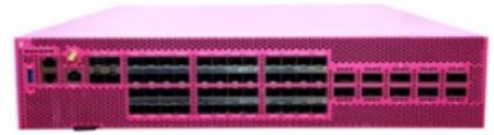
ufiSpace

Telefonica

Infinera

Edge-corE NETWORKS

Virtual Broadband Network Gateway

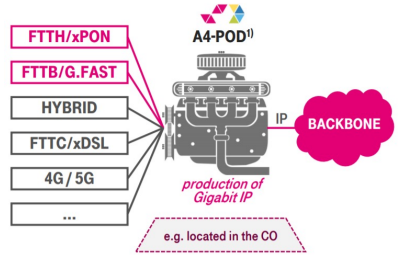


Qumran2c

Deutsche Telekom

rtbrick

DELTA

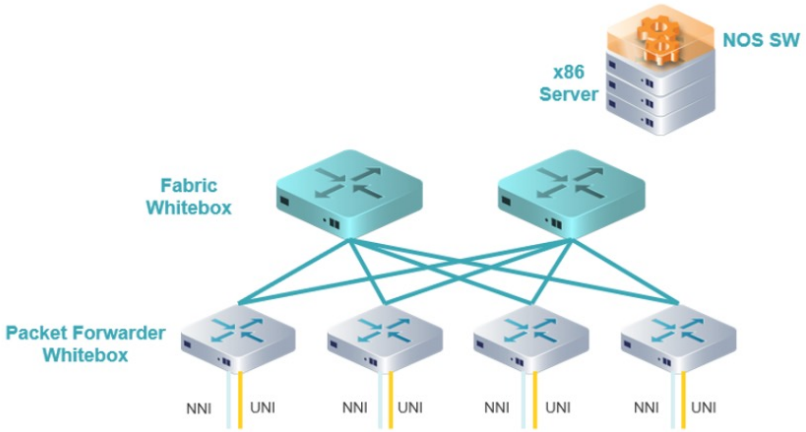


TIP Distributed Disaggregated Backbone Router (DDBR)

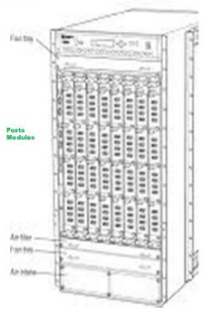
KDDI

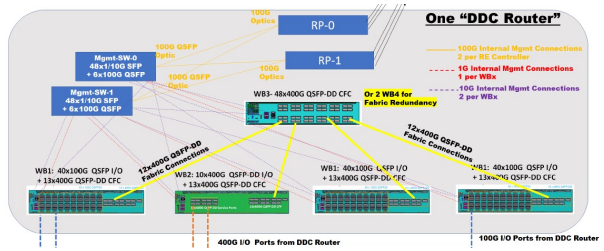
DELTA

DRIVENETS



Core & Edge Router via DDC





Jericho2

Ramon

ufiSpace

DRIVENETS

cisco

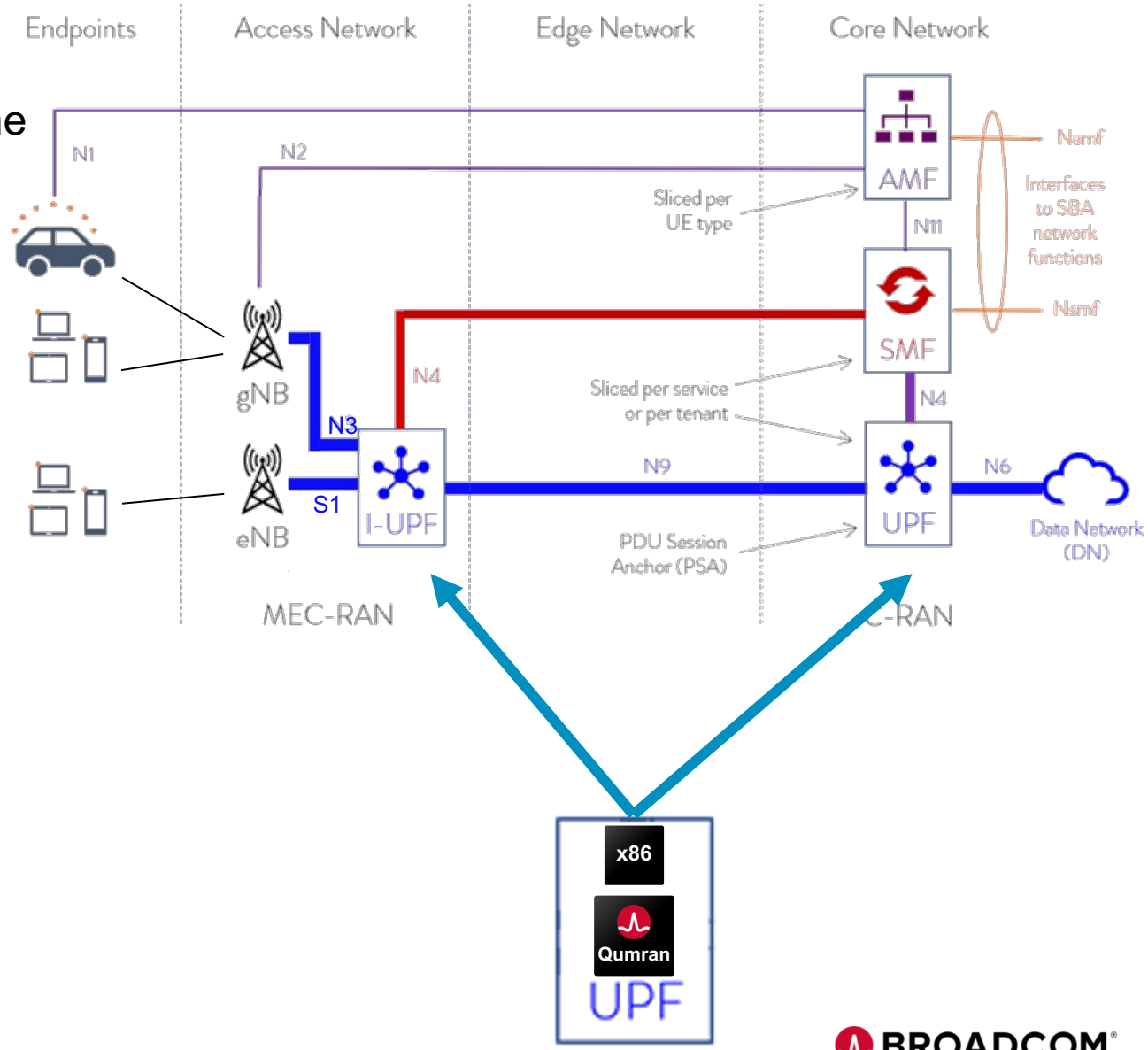
OPEN Compute Project

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Qumran-Based Router Integrating UPF

- Single-chip Session, Forwarding and Routing plane
  - GTP-U processing
  - Carrier-Grade Segment Routing
  - Network Slicing
  - HQoS and packet buffering
  - Routing, ACL
  - Any tunneling technology (TE, SR, etc.)
  - Advanced telemetry, statistics and Instrumentation
  - Future-proofing programmability
- CUPS and Disaggregation ready model
- Significant cost and power saving



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JR2/J2C+ Series GTP-U Service Scale in Centralized Mode

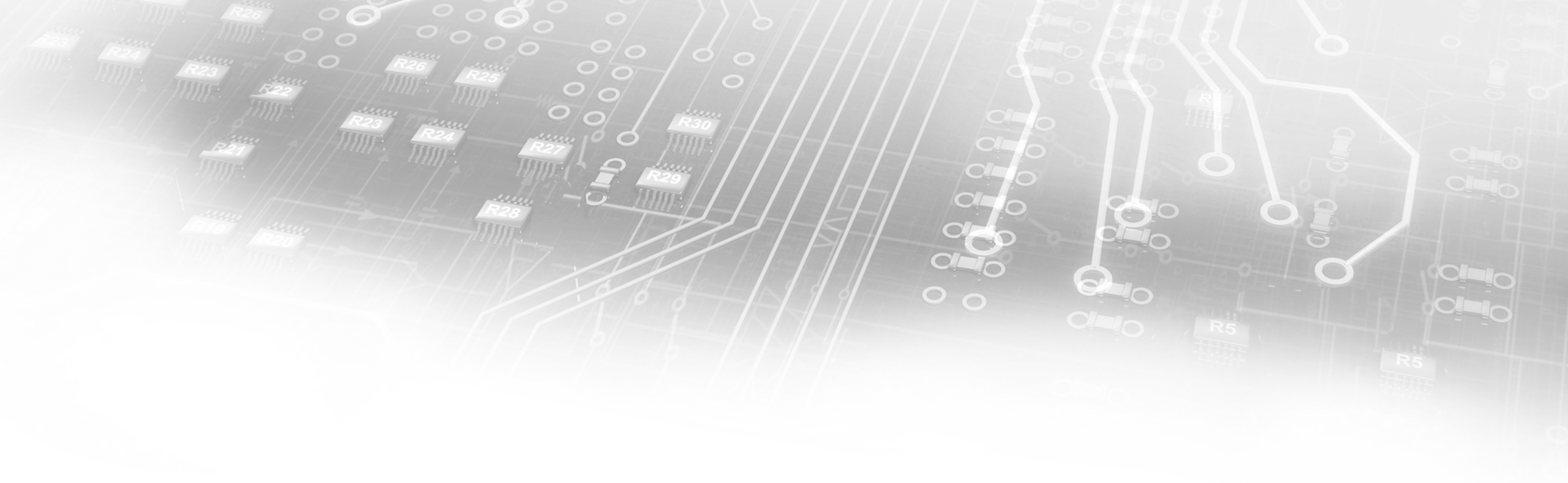
	Q2c+	JR2	Q2c	J2x	Q2a	Q2u	Q2n
N3-Facing Interfaces (no over-subscription)	3.6Tbps	2.4Tbps	1.2Tbps	800Gbps	400Gbps	180Gbps	40Gbps
New Connection Rate	100-150K/s	100-150K/s	100-150K/s	100-150K/s	50-75K/s	50-75K/s	25-40K/s
Fixed Wireless Service Scale *	500K	120K	120K	120K	30K	30K	30K
Mobility Service Scale *	1M	320K	300K	300K	100K	100K	100K

- Fixed Wireless Access (FWA)
  - Min bandwidth requirement of 100Mb/s per UE (10K UE / 1Tbps)
  - Upstream metering and downstream shaping per UE
  - VOQ and counter per UE, with smart counter eviction
- Mobility
  - Min bandwidth requirement of 10Mb/s per UE (100K UE / 1Tbps)
  - On demand fine grained Upstream metering and downstream shaping, or per UE group (or service type)
  - On demand fine grained VOQ or per UE group (or service type)
  - Counter per UE, with smart counter eviction

* 2-hop processing where applicable

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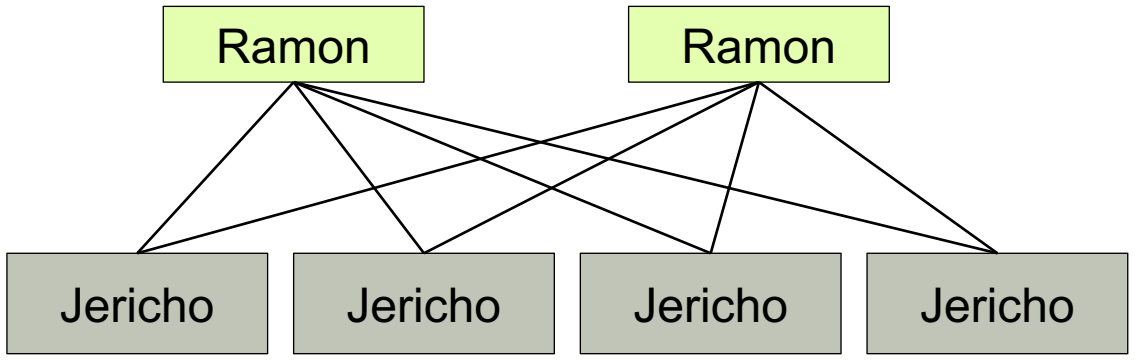


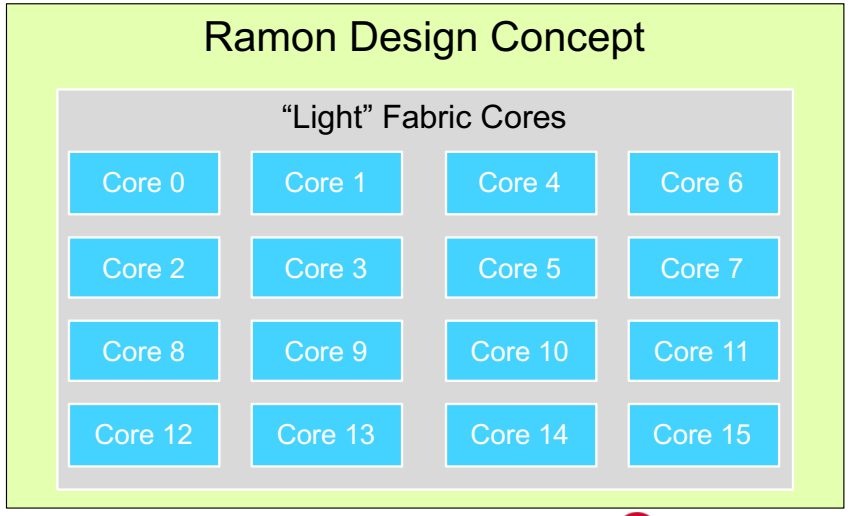
DNX Architecture Efficiency vs. Ethernet

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Power Efficient Architecture - Ramon

- Ramon is the industry most power efficient switch
- 25% power saving compared to best of class equivalent Ethernet switch
- Ramon is used as a high-performance cell switching fabric device, Zero design overhead
  - Optimized design
  - Doesn't carry multipurpose logic targeted to reduce R&D spending over multiple tape outs
  - Light, elegant and proven design
  - Fine-tuned for multiple generations





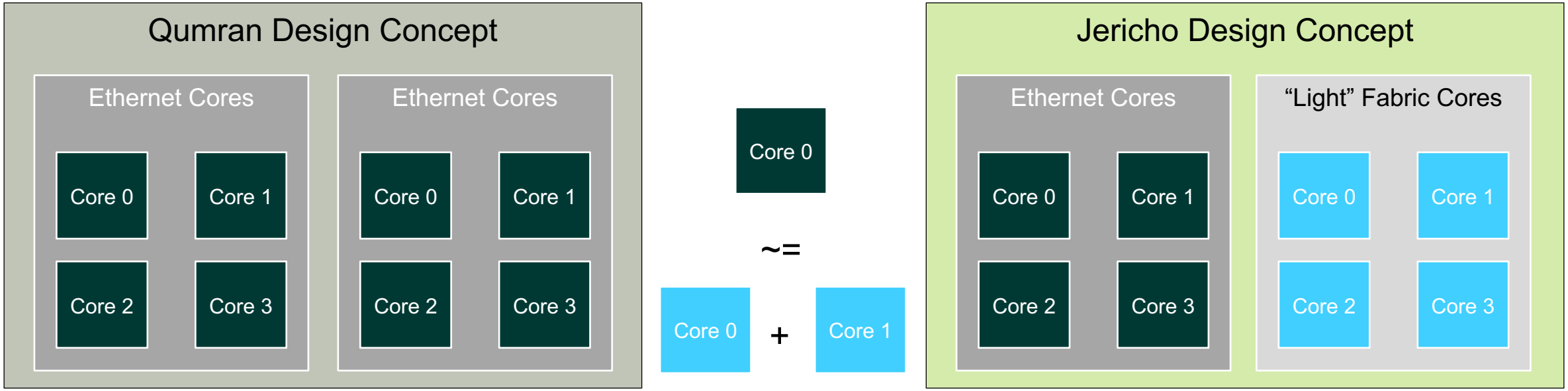
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Power Efficient Architecture – Jericho

- Same “zero design overhead” concept
- Optimized for line-card / NCP functionaility, no combo functionaility overheads
- Light and optimized fabic cores reduce total device power
- No fabric logic overhead on NIF cores

Power, Ethernet Cores X2 vs. Fabric

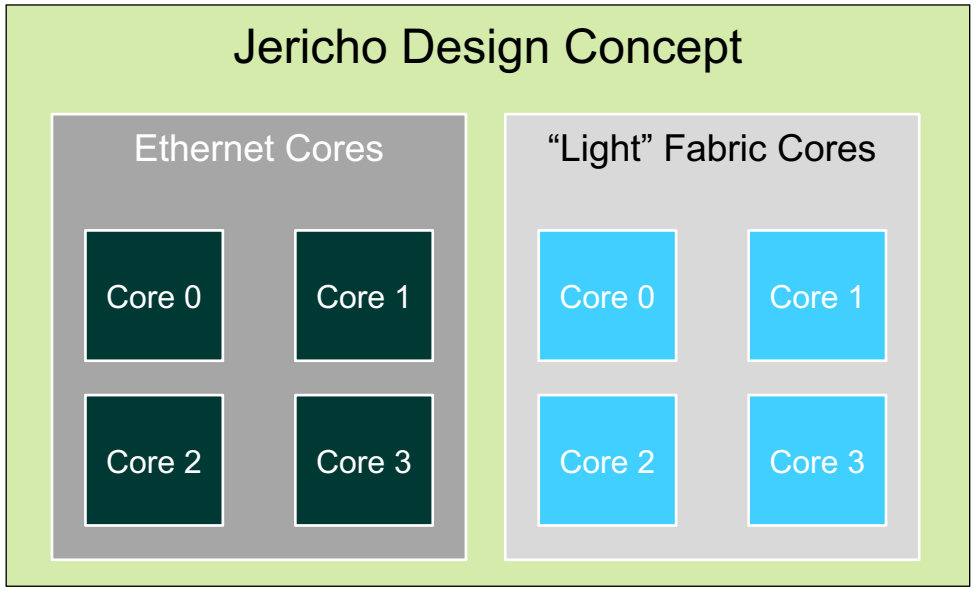


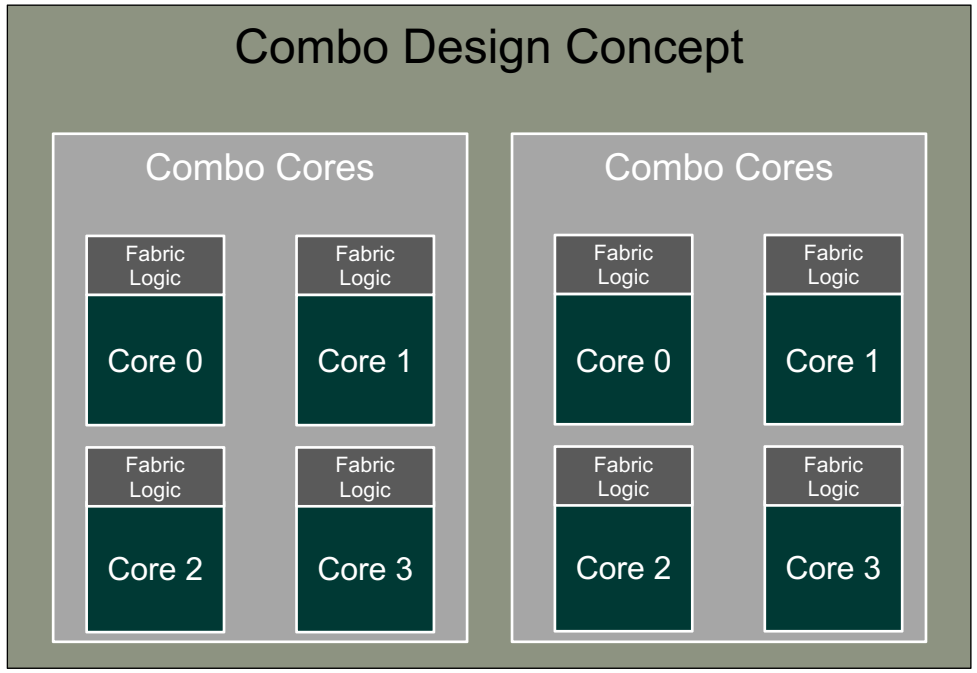
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Combo Arch – “One arch fits all”

- Each core can act as fabric or Ethernet
- Generic engines carry power overhead vs. optimized logic





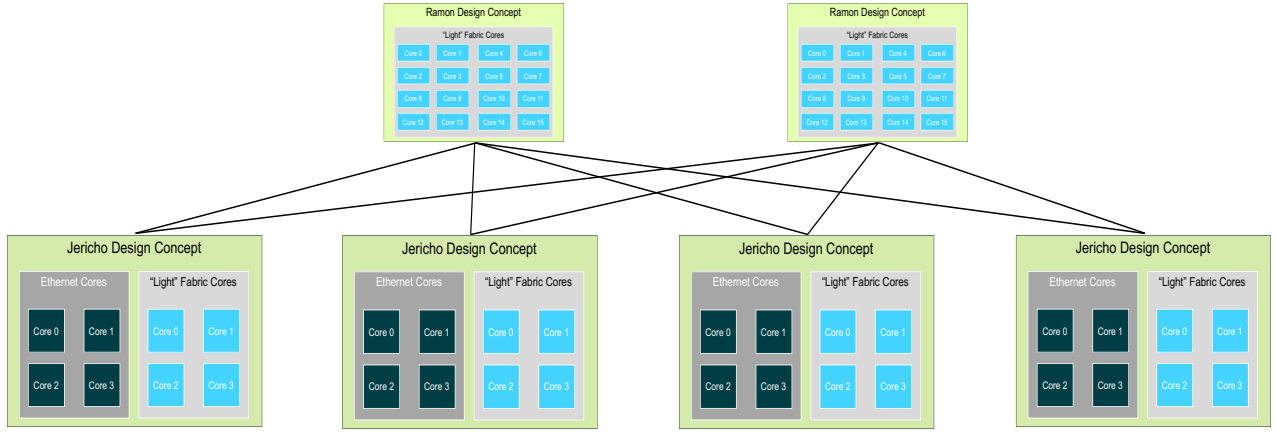
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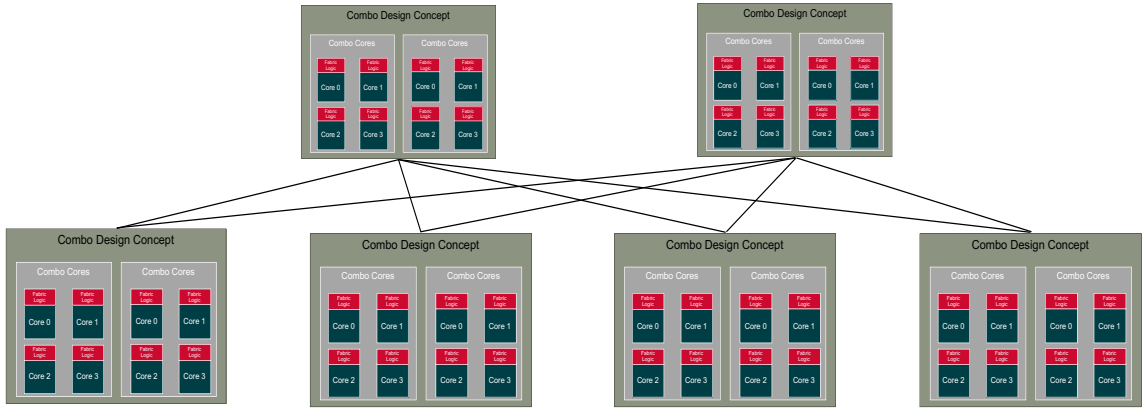
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Network View – What Do You Optimize For?

- Zero design overhead
- Optimized for low power

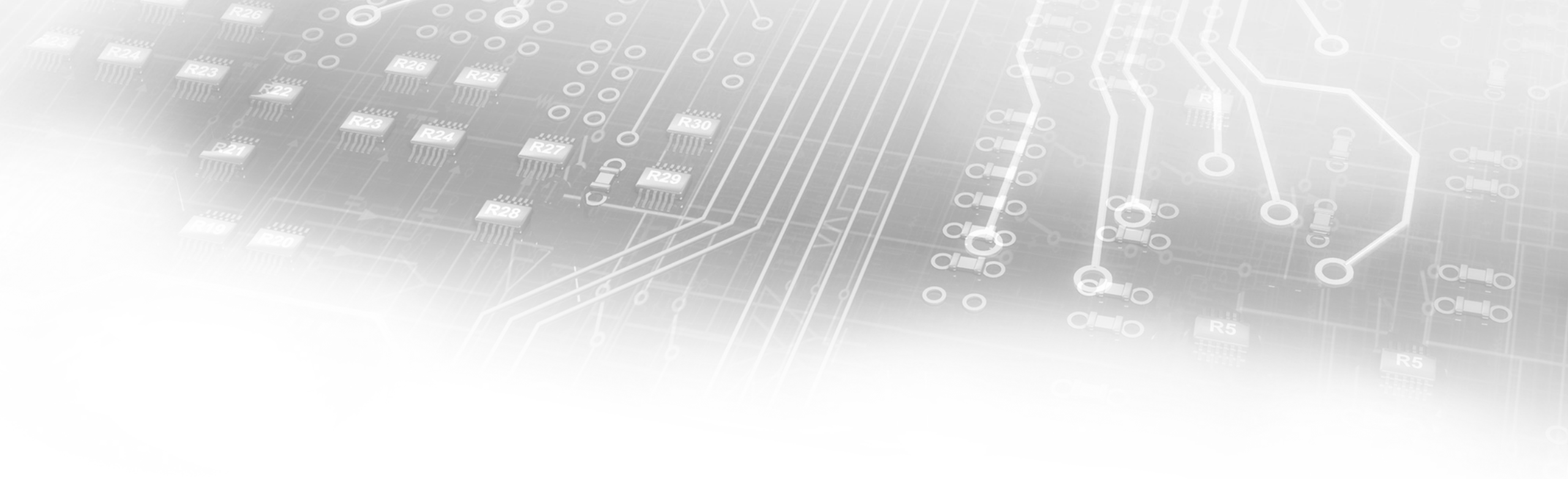
- One device fits all
- Optimized to reduce R&D resources





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DNX DDC for AI / ML

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AI Network Workload

Requirements

- Objective
  - Reduce job completion time
  - Increase GPU utilization
- Characteristics
  - High network utilization
  - Small Number of flows at high bandwidth
  - Uneven bandwidth flows

Network Pain Points

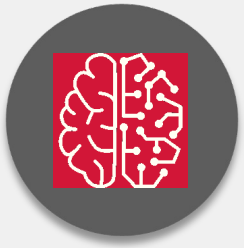
- Standard Load balancing is flow-based
- Leads to
  - Uneven link utilization
  - Sub-optimal network utilization
  - Frequent Congestion



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The Distributed Disaggregated Chassis (DDC)

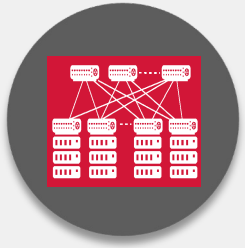


Native for AI/ML network

- Perfect load balancing
- Flat network - single routing domain
- E2E scheduling
- No congestion
- Scalable and high radix



With DNX



Industry-largest router

- Disaggregated solution
- Yielding the industry-largest single routing domain
- Logically-equivalent to vertically integrated solutions



Wide, open eco-system

- AT&T first to deploy at the core network
- Adopted by leading carriers and cloud providers
- OEM and 3rd party software over Jericho and Ramon fixed boxes

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DDC White Box Ecosystem

In Deployment

- DDC scale-out to 18,000 400GE ports
- Network Cloud Processor 3 (Jericho2c+)
  - 40x400G fabric ports
  - 36x400GE network ports
- Network Cloud Fabric (NCF, Ramon)
  - 48x400G fabric ports

2023

- DDC scale-out to 32,000 800GE ports
- Jericho3 NCP
  - 18 x 800GE / 36 x 400GE network ports
  - 20 x 800G fabric ports
- Ramon3 NCF
  - 64 x 800G fabric ports

NCP2 8T J2 16nm
500W Max power



10x400GE +13x400G

NCF 9.6T Ramon 16nm



48x400G

NCP3 28.8T J2c+ 7nm
1000W Max power



36x400GE + 40x400G

NCP4 28.8T J3 5nm
600W Max power



36x400GE + 20x800G

NCF2 51.2T R3 5nm



64x800G

NCP in Production

2019 2021 2023

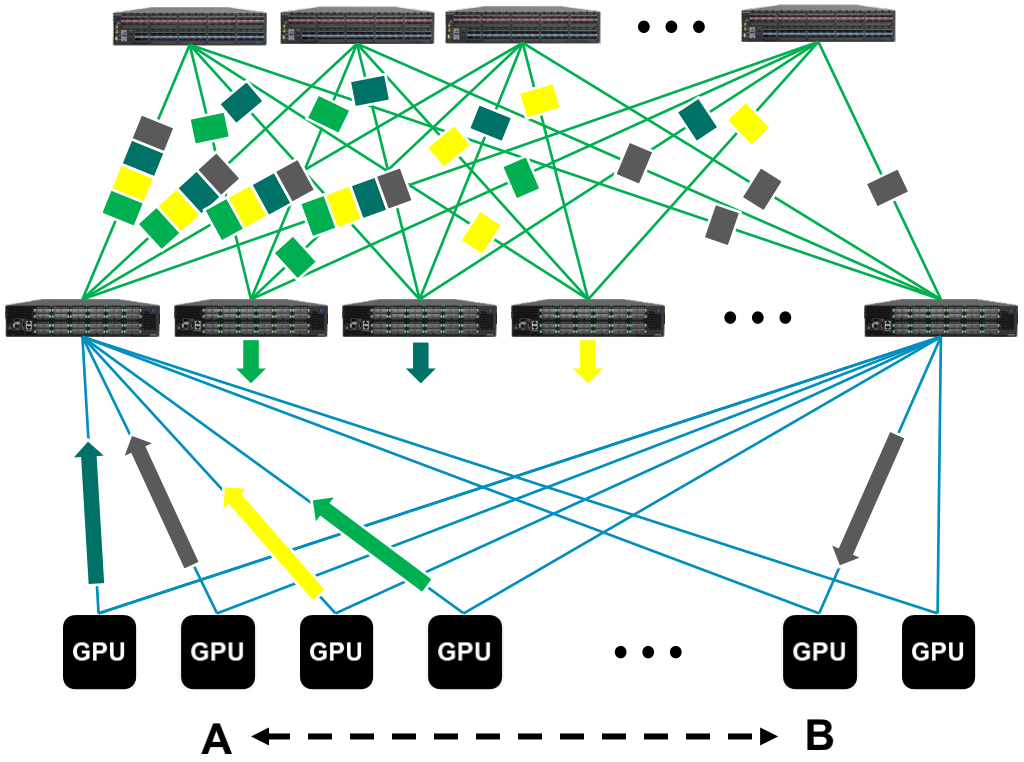
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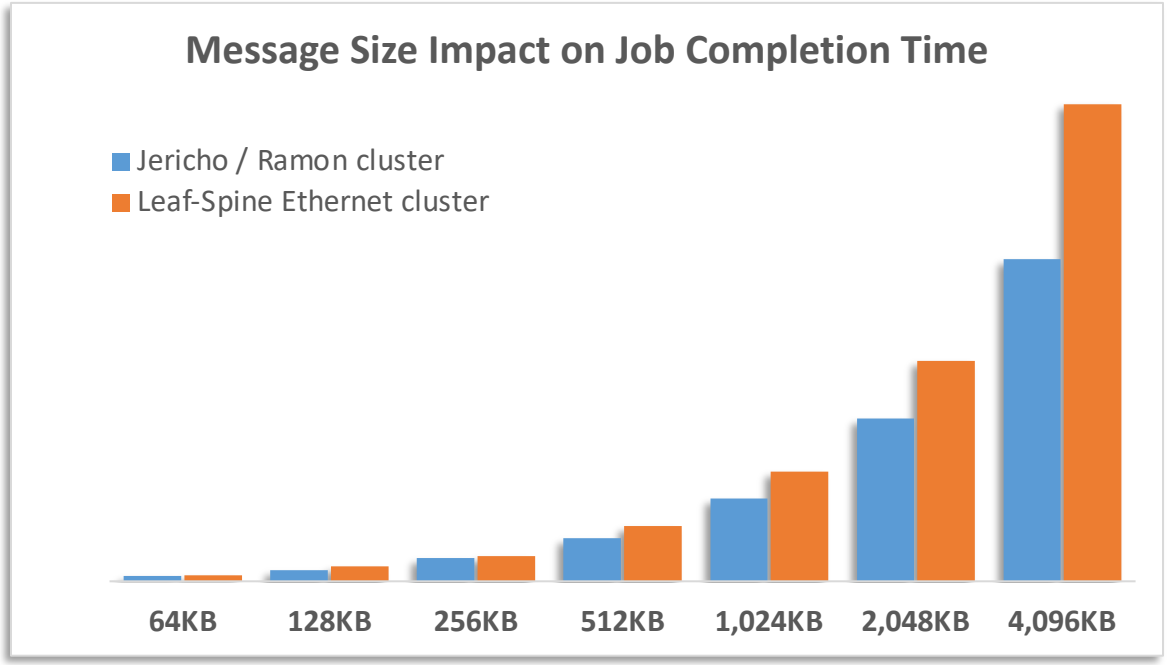
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Reducing Job Completion Time with DNX DDC

Perfect load balancing

- Larger messages in certain flows can overload the links
- DNX fabric uses all links for every flow – cell spraying





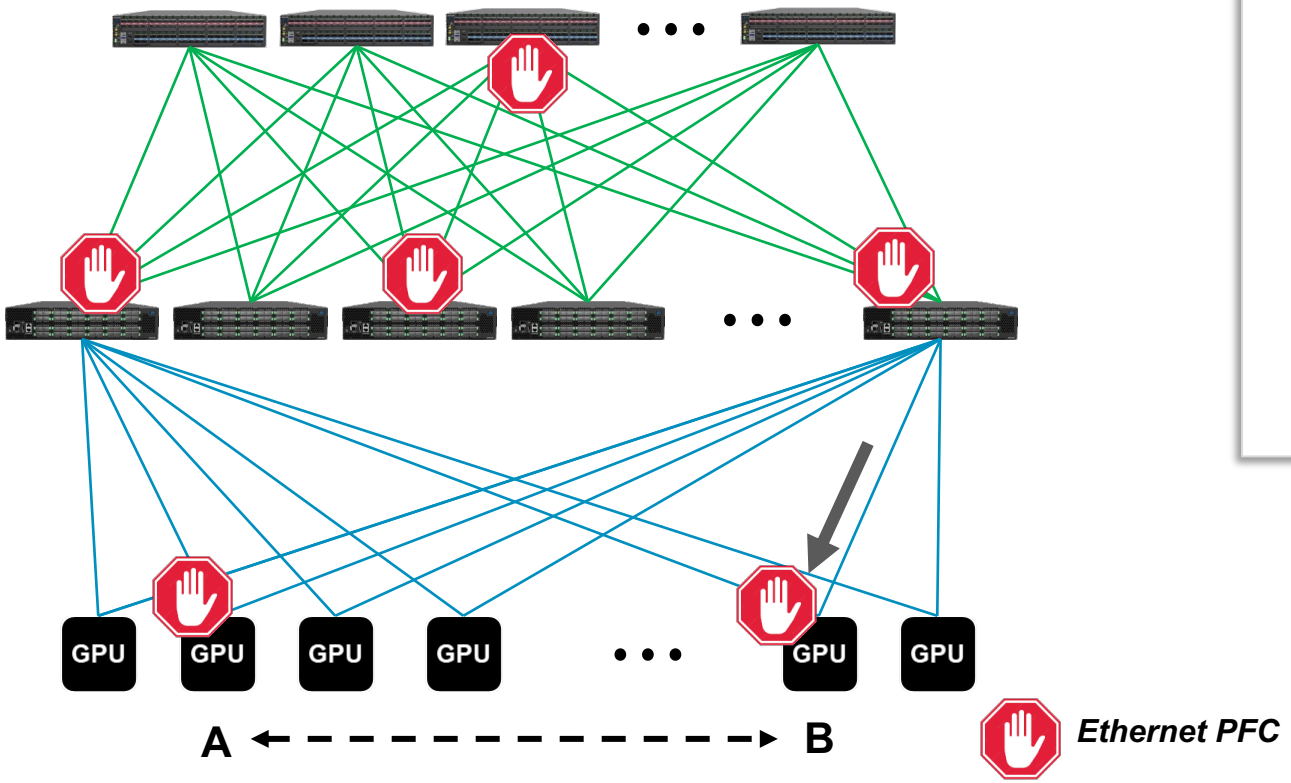
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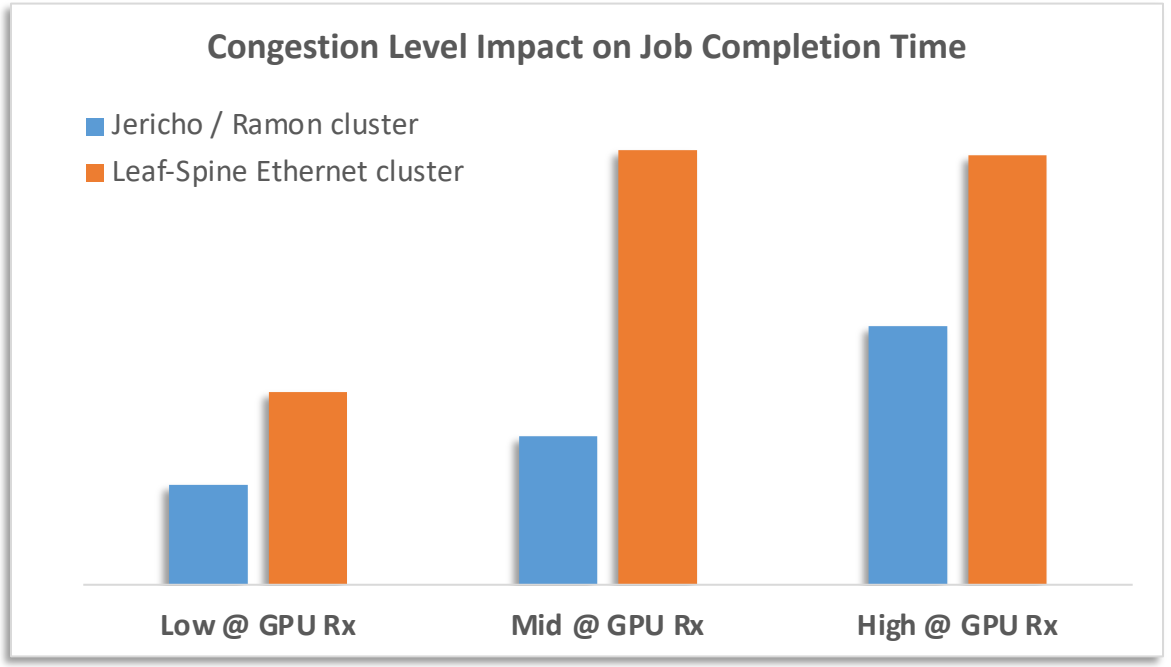
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Reducing Job Completion Time with DNX DDC

End-to-End Congestion Control

- Ethernet PFC propagates to all links, creates HOL blocking
- DNX E2E scheduling eliminates HOL blocking
  - Flow held at the source





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Summary – DDC in the AI/ML Network



Perfect load balancing
Cell spraying over the cell-based DNX fabric
No hashing, no ECMP polarization

End-to-end scheduling
VOQ-based

Lossless RDMA operation
Deep buffering and Virtual output queuing

Low and controlled latency

Single-tier, large-scale routing domain for the entire network
Simplified management

- Further reading
  - “Distributed Decoupling Chassis” - white paper (in Chinese) by China Telecom and Broadcom
  - “Reducing job completion time in AI/ML clusters” – Broadcom blog by Xiguang (Henry) Wu

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Summary

- DNX is the leading switch/routing technology, deployed in the world's largest carriers and MSDCs
- Mature and field-proven SDK and architecture (10th generation)
- Extensive and robust portfolio with committed investment and cadence
- Vibrant hardware & Software ecosystem enabling multiple solutions to the end customer
- Access, BNG, Edge, Peering, and Core rich feature-set, all deployed in T1 carrier networks

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Thank You



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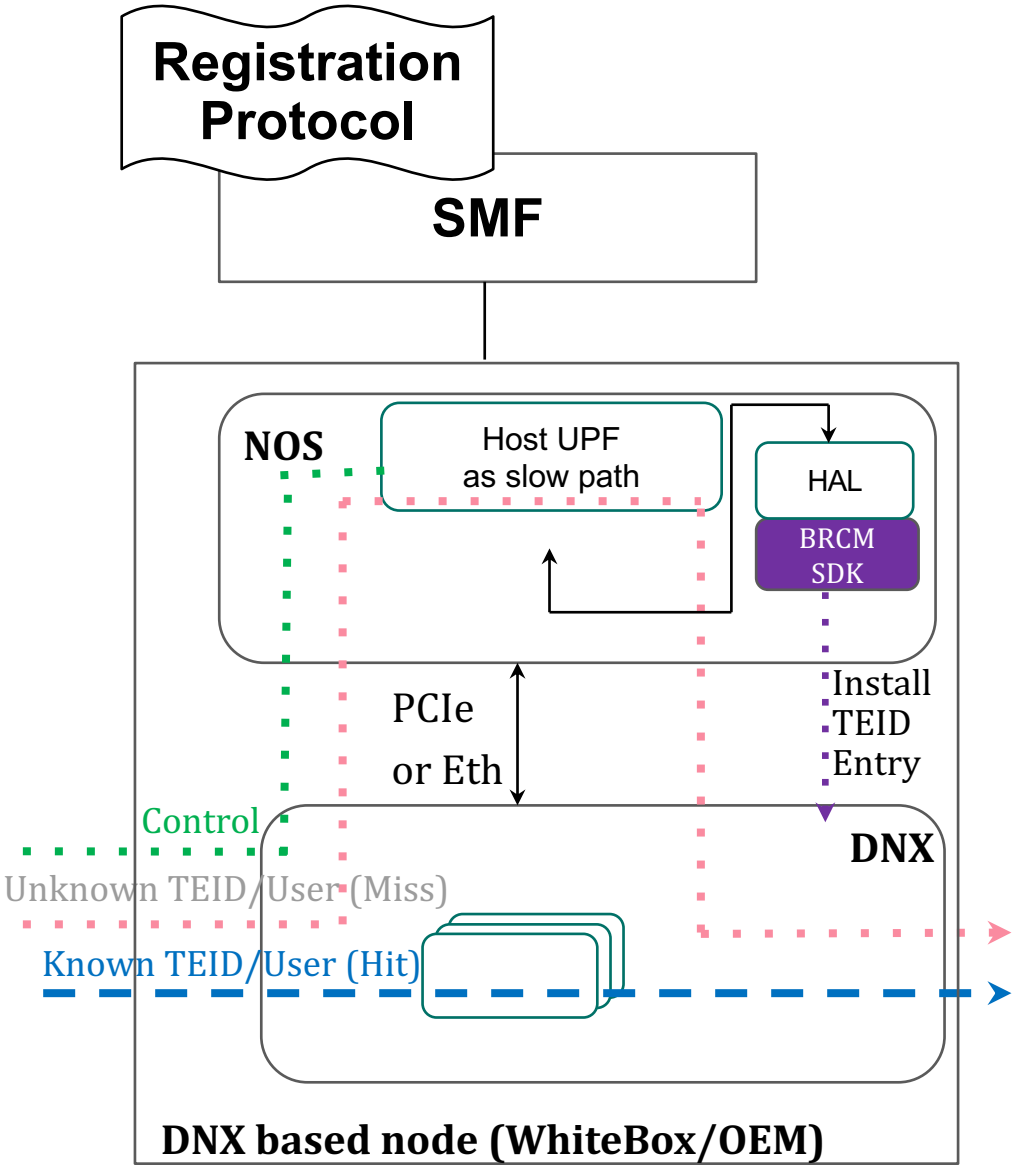
backup

November 1, 2022

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UPF Node – Example

- UPF scales from 80Gbps to 7.2Tbps, from 10GE to 400GE
- Qumran offloading the UPF datapath
  - Termination/encapsulation the GTPUv1 PDU with the PDU Container(0x85) extension header according to 3GPP
  - Same forwarding (Match-Action table) implemented in UPF data-path
  - MEC Servers aggregation Router
  - QoS
  - DPI redirect
- Switch Host CPU
  - NOS run exactly same UPF function as current X86 UPF does
  - Can act as slow path (network-defined)
  - Extend the NOS HAL to populate data-path flow tables in Qumran
- GTP Control Redirect to Slow-path
  - Includes: GTPCv1, GTPCv2, GTP', GTPUv1-signaling

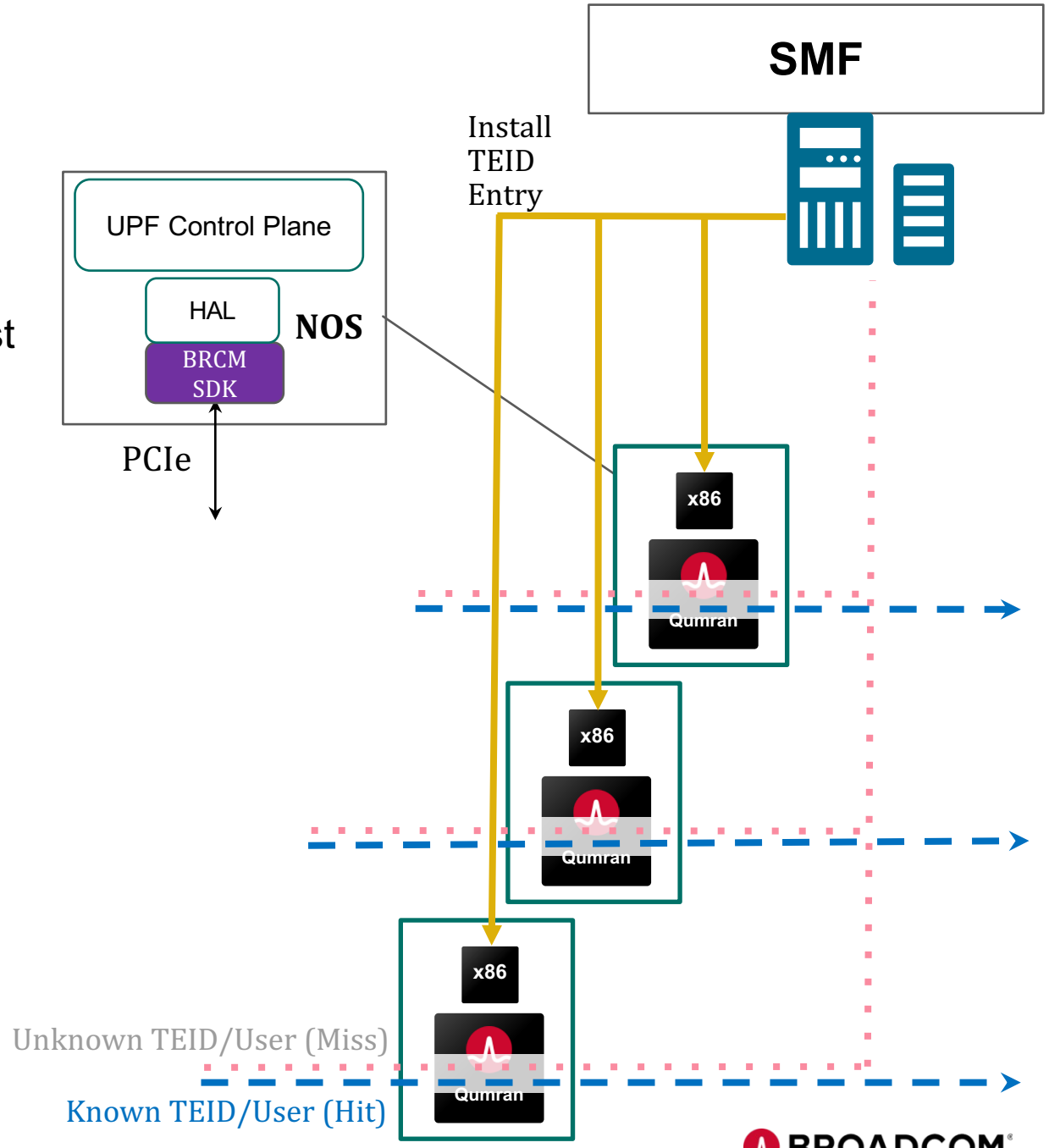


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Cloudifying The Slow Path

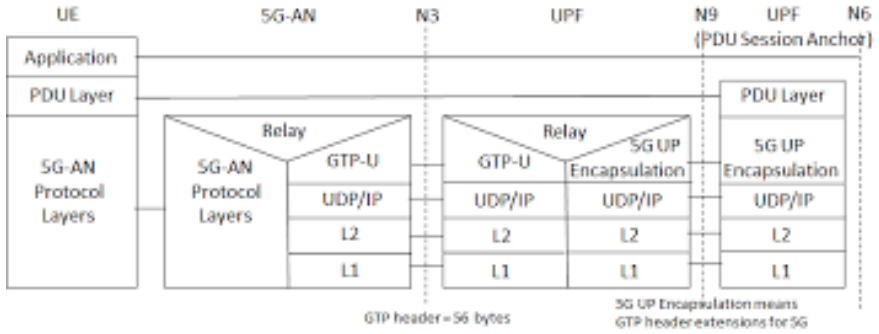
- Unknown TEID sent to a remote “slow path” service
- New TEID configuration provided to the local x86 host
- Local host installs the TEID in Qumran



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UPF Market Trends



GTP Tunnels

DNX

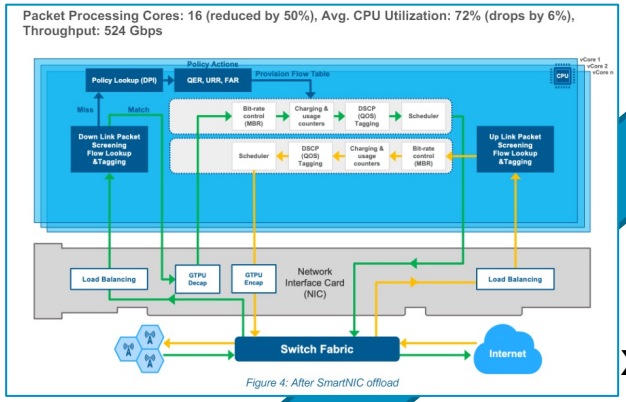


Figure 4: After SmartNIC offload

x86 + SmartNIC



x86

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Jericho2 Series UPF: GTP-U Released!

In Trials with a Chinese Carrier

UPF-phase 1
Now

- Control plane trap
- GTPUv1 Encapsulation and Termination
- GTPUv1 \w Session Container Termination
- Up to 200K* TEID/User-IP Scale
- Counters
- Upstream Policing + Downstream Shaping
- FWA Use Case
- Mobility Use Case
- Lawful intercept
- N6 Tunneling: MPLS, IP, SR, etc.

UPF-phase 2
Q4 2021

New features

- GTPUv1 \w Session Container Encapsulation
- QoS Mapping (QFI)
- Up to 0.5M-1M* TEID/User-IP Scale on-chip

* Jericho2 scale

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Released Segment Routing Techniques

	"Classic" SRv6	uSID	G-SID	MPLS-SR
Availability	**DNX SDK in GA**	**DNX SDK in GA**	**DNX SDK in GA**	**DNX SDK in GA**
Compression	Full 128b, no compression	32b prefix, 16b uSID (48b prefix in the plan)	48b prefix, 32b G-SID	N/A
JR2 max. scale	Up to 11 SID	Up to 66 uSID	Up to 16 G-SIDs (Roadmap to increase in J2c+)	Up to 14 labels
VPN (mix) example	1 x VPN + 10 x SID	1 x "classic" (VPN) + 60 x uSID	1 x "classic" (VPN) + 12 x G-SID	N/A
Deployment Status	Deployed	Deployed	Interop by CMCC JR2 with ZTE, Huawei, H3C	Deployed
Standard	spring-srv6-network-programming-24	spring-net-pgm-extension-srv6-usid-07	generalized-srv6-for-cmpr-01	draft-ietf-spring-segment-routing-mpls-22

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